Discussion Exercise 4

University of California, Riverside
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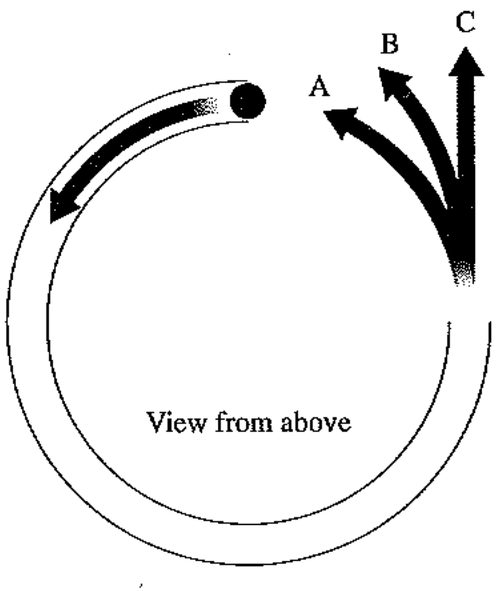
1. Equilibrium

A hollow tube forms three-quarters of a circle. It is lying flat on a table. A ball is shot through the tube at high speed. As the ball emerges from the other end, does it follow path A, path B or path C? Explain your reasoning.

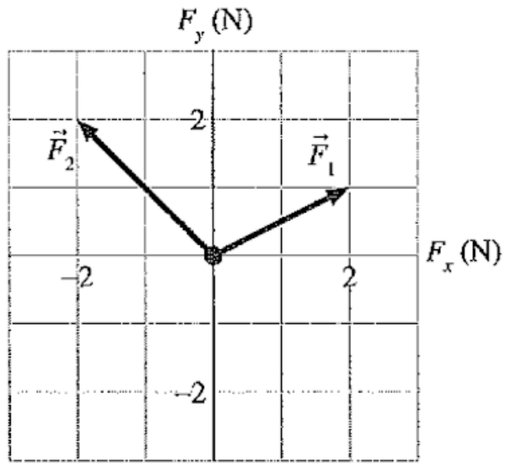


C. *Since the net force applied to the ball outside the tube should be zero without considering any frictions, then by Newton's first law, it moves at constant velocity.*

2. Dynamics

A Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ cause a 1 kg object to accelerate with the given acceleration. Two of the forces are shown on the free-body diagrams below, but the third is missing. For each, draw and label on the grid the missing force vector.

$a_x = 2\ \mathrm{m/s^2}$
$a_y = 0\ \mathrm{m/s^2}$



Solving numerical problems in physics

- Understand the problem and set up a model
  - Describe/explain in your own words
- Visualize
  - Draw the diagrams
  - Label relevant physical quantities (known and unknown)
- Identify the physics concepts
- Find the relationship between known and unknown quantities
  - Derive the symbolic expressions (Calculate numbers at the last step)
  - Check units
- Review the results
  - Reasonable values?
  - Correct units?

2. Dynamics

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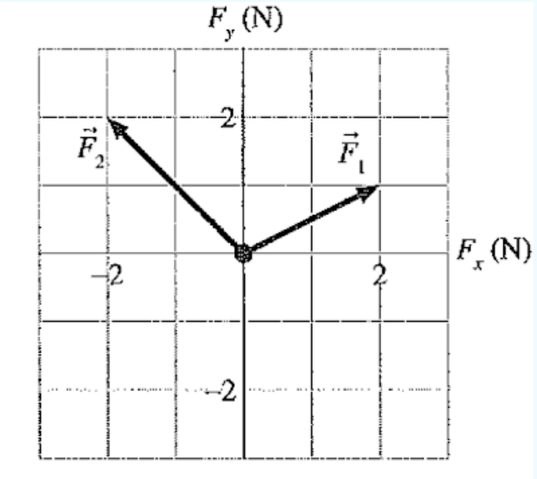
Visualization

$a_x = 2\ \text{m/s}^2$
$a_y = 0\ \text{m/s}^2$



Physics concepts (Newton's second law:)

$$\vec{F} = m\vec{a}$$

1. $F_{1x} + F_{2x} + F_{3x} = ma_x$
2. $F_{1y} + F_{2y} + F_{3y} = ma_y$

Known

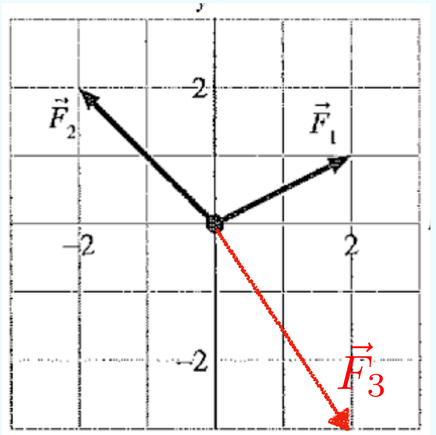
$F_{1x} = 2$
$F_{1y} = 1$
$F_{2x} = -2$
$F_{2y} = 2$
$m = 1kg$
$a_x = 2m/s^2$
$a_y = 0m/s^2$

Find

$F_{3x} = ?$
$F_{3y} = ?$

1. $2 + (-2) + F_{3x} = 1 \times 2$
2. $1 + 2 + F_{3y} = 0$

$F_{3x} = 2$
$F_{3y} = -3$



Visualization → Physics concepts → Equations: Unknown ⟷ Known

2. Dynamics

A Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ cause a 1 kg object to accelerate with the given acceleration. Two of the forces are shown on the free-body diagrams below, but the third is missing. For each, draw and label on the grid the missing force vector.

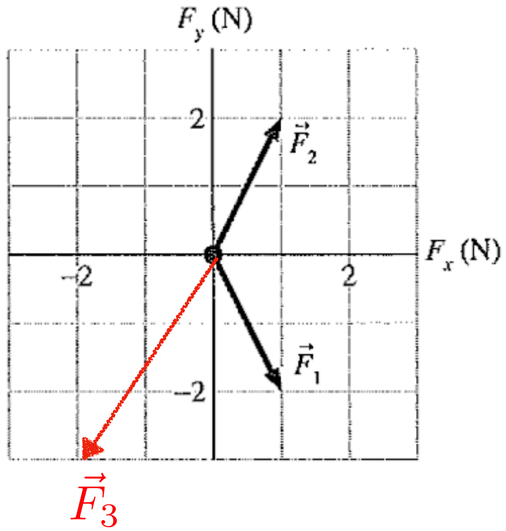
Newton's second law:

$$F_{1x} + F_{2x} + F_{3x} = ma_x$$
$$F_{1y} + F_{2y} + F_{3y} = ma_y$$

$a_x = 0 \text{ m/s}^2$
$a_y = -3 \text{ m/s}^2$

1. $1 + 1 + F_{3x} = 0$
2. $2 + (-2) + F_{3y} = 1 \times (-3)$

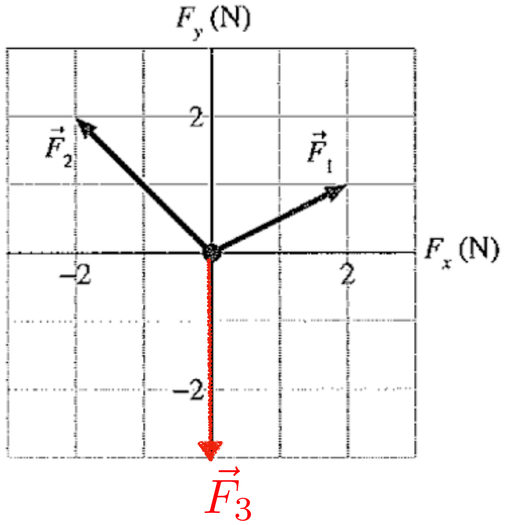
$F_{3x} = -2$
$F_{3y} = -3$



The object moves with constant velocity.

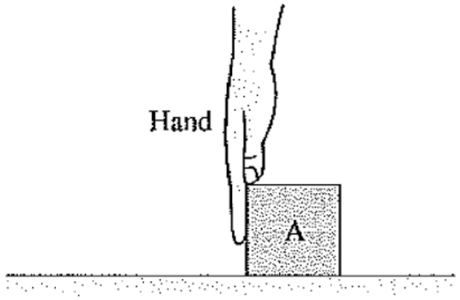
1. $2 + (-2) + F_{3x} = 0$
2. $1 + 2 + F_{3y} = 0$

$F_{3x} = 0$
$F_{3y} = -3$

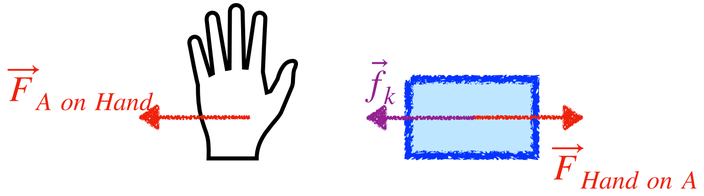


3. Interacting Objects

25. Block A is pushed across a horizontal surface at a *constant* speed by a hand that exerts force $\vec{F}_{\text{H on A}}$. The surface has friction.



a. Draw two free-body diagrams, one for the hand and the other for the block. On these diagrams, show only the *horizontal* forces with lengths portraying the relative magnitudes of the forces. Label force vectors, using the form $\vec{F}_{\text{C on D}}$. On the hand diagram, show only $\vec{F}_{\text{H on A}}$. Don't include $\vec{F}_{\text{body on H}}$.



b. Rank in order, from largest to smallest, the magnitudes of *all* of the horizontal forces you showed in part a. For example, if $F_{\text{C on D}}$ is the largest of three forces while $F_{\text{D on C}}$ and $F_{\text{D on E}}$ are smaller but equal, you can record this as $F_{\text{C on D}} > F_{\text{D on C}} = F_{\text{D on E}}$.

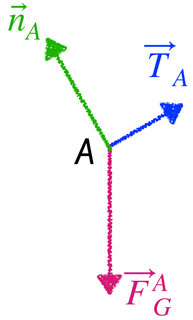
Order: $F_{Hand\ on\ A} = f_k = F_{A\ on\ Hand}$

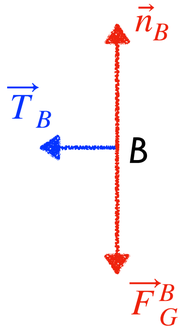
Explanation: $F_{Hand\ on\ A} = F_{A\ on\ Hand}$ since they are action/reaction pair.

$F_{Hand\ on\ A} = f_k$ since the acceleration is zero, then the net force should be zero as well.

4. Ropes and Pulleys

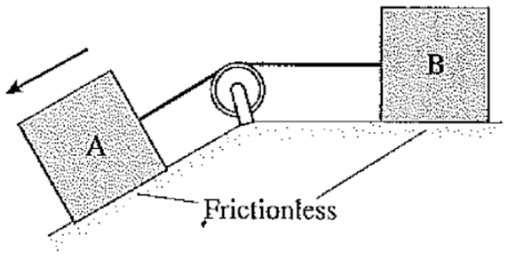
Exercises 31–32: Draw separate free-body diagrams for blocks A and B. Indicate any pairs of forces having the same magnitude. Assume $m_A \neq m_B$

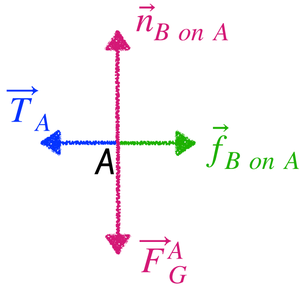


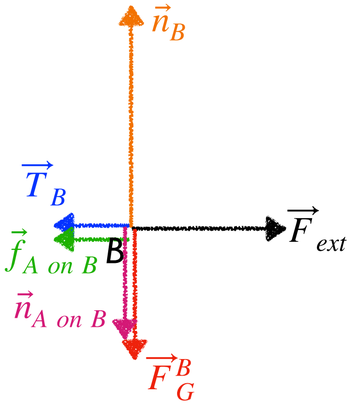


$T_A = T_B$ since the tension force in the massless rope should be the same everywhere.

$F_G^B = n_B$ since there is no acceleration on y-direction for B.



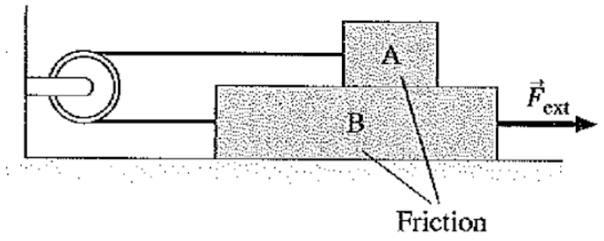




$T_A = T_B$ since the tension force in the massless rope should be the same everywhere.

$n_{B\ on\ A} = F_G^A$ since there is no acceleration on y-direction for B.

$n_{B\ on\ A} = n_{A\ on\ B}$, $f_{B\ on\ A} = f_{A\ on\ B}$ since they are action/reaction pairs.



If there is no relative motion between A and B, then A keeps stationary, so we have one more equality: $T_A = f_{B\ on\ A}$